Dr. Thomas Langerak
HCI + AI Postdoctoral Researcher
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RESEARCH INTEREST

My research develops a learning-based foundation for intelligent interaction, bridging human–computer interaction and artificial intelligence. I model users as adaptive decision-makers and design algorithms that regulate autonomy and feedback between humans and machines. This work combines reinforcement learning, optimization, and interactive systems, unifying sensing, actuation, and decision-making. As interfaces dissolve in the age of intelligent agents, my goal is to make the adaptive strategy itself the new interface.

ACADEMIC POSITIONS

Postdoctoral Researcher *10/2024 - Present*
Human-Computer Interaction & Artificial Intelligence
Computational Behavior Lab, Aalto University, Finland.
Advisor: Prof. Antti Oulasvirta

- Developing Reinforcement Learning based approaches for modeling human decision making under partial observability.
- Researching methods for cost-aware Bayesian optimization for design and prototyping that better fit the design process.
- Teaching courses (Master and Ph.D. level) and student supervision (interns and M.Sc. thesis).

EDUCATION

Ph.D. Student *10/2018 - 07/2024*
Human-Computer Interaction & Artificial Intelligence
Advanced Interactive Technologies, ETH Zürich, Switzerland.
Advisors: Prof. Otmar Hilliges & Prof. Christian Holz
Dissertation: User Agency and System Automation in Interactive Intelligent Systems
Committee: Prof. Christian Holz, Prof. Antti Oulasvirta, and Prof. Roderick Murray-Smith

- Multi-agent reinforcement learning to simultaneously learn a user model and an adaptive policy to maximize user performance.
- Model-predictive control for haptic devices that incorporates predictive models of human behavior for agency-automation trade-off.
- Developing novel actuators and sensing algorithms for haptic devices.
- Teaching (B.Sc., M.Sc., and Ph.D.) as well as student supervision.

Double-degree M.Sc. Student *09/2016 - 08/2018*
Computer Science (Intelligent Systems)
Aalto University & Twente University.

- With Honors (Distinction) from Aalto University
- Courses in Machine Learning, HCI, User Interfaces, Design, and multi-agent systems.
- Minor: Entrepeneurship

Exchange Student (Sophomore Year) *01/2015-06/2015*
Industrial Design
Carnegie Mellon University, USA

- Semester Exchange to Carnegie Mellon University.
- Focus on product and computational design.

B.Sc. Student *09/2012 - 02/2016*
Industrial Design
Eindhoven University of Technology, the Netherlands

- Specializing in tangible and embodied interaction.
- Focus on software and electrical engineering.

PROFESSIONAL EXPERIENCE

Research Scientist Intern *05/2023-10/2023*
Human–Computer Interaction & Explainable AI
Meta Reality Labs, Redmond, USA

- Bayesian user models for better explainable AI in the context of adaptive interfaces.
- Completed a full research project to successful publication ([J.1.]).

Intern *09/2014-01/2015*
Industrial Design & Prototyping
Studio Sophisti, Amsterdam, the Netherlands

- Prototyping, Software, Electrical Engineering and Design for interactive devices.
- First iteration on a Core77 design award winner .
- Worked for Lego, Disney and Hasbro.

PUBLICATIONS

Under Review & Ongoing

[WIP.1]: *Bayesian Optimization for Prototyping Interactive Devices*
Thomas Langerak, Renate Zhang, Morgan Wang, Per Ola Kristensson and Antti Oulasvirta. 2025. Under Review.

[WIP.2]: *Design Optimization with Multiple Evaluation Sources using Multi-Surrogate Bayesian Optimization*
Francisco Fernandes, **Thomas Langerak**, Mira Kernen, Danqing Shi, Ardak Alipova, and Antti Oulasvirta. 2025. Under Review.

[WIP.3]: *RILe: Reinforced Imitation Learning.*
Mert Albaba, Sammy Christen, Christoph Gebhardt, **Thomas Langerak**, Michael J. Black, and Otmar Hilliges. 2024. Under Review.

[WIP.4]: *Inverting POMDPs with Unknown Observation Functions and Rewards.* In Progress.

[WIP.5]: *Learned Latent Rational Control for Modeling Human Gaze Behavior.* In Progress.

Conference Papers

[C.1]: *MARLUI: Multi-Agent Reinforcement Learning for Goal-Agnostic Adaptive UIs.*
Thomas Langerak, Sammy Christen, Mert Albaba, Christoph Gebhardt and Otmar Hilliges. 2024. Proc. ACM Hum.- Comput. Interact. 8.

[C.2]: *Hedgehog: Handheld Spherical Pin Array based on a Central Electromagnetic Actuator.*
Aline Abler, Juan Zarate, **Thomas Langerak**, Velko Vechev and Otmar Hilliges. 2021. In World Haptics Conference.
Honorable Mention

[C.3]: *Omni: Volumetric Sensing and Actuation of Passive Magnetic Tools for Dynamic Haptic Feedback.*
Thomas Langerak*, Juan Zarate*, David Lindlbauer, Christian Holz, and Otmar Hilliges. 2020. In Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology.

[C.4]: *Optimal Control for Electromagnetic Haptic Guidance Systems.*
Thomas Langerak, Juan Zarate, Velko Vechev, David Lindlbauer, Daniele Panozzo, and Otmar Hilliges. 2020. In Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology.

[C.5]: *Contact-free Nonplanar Haptics with a Spherical Electromagnet.*
Juan Zarate*, **Thomas Langerak***, Bernhard Thomaszewski and Otmar Hilliges. 2020. IEEE Haptics Symposium.

Journal Papers

[J.1]: *XAIUI: Mental-Model Aware Explainable AI for Adaptive Interfaces.*
Thomas Langerak, Kashyap Todi, Ben Lafreniere, Ruta Desai, and Tanya Jonker. 2025. Transactions on Interactive Intelligent Systems.

[J.2]: *Robust Real-Time Tracking of Axis-Symmetric Magnets via Neural Networks.*
Mengfan Wu*, **Thomas Langerak***, Juan Zarate and Otmar Hilliges. 2024. IEEE Transactions on Magnetics.

Auxiliary: Demos, Posters, and Workshops

[A.1]: *RILe: Reinforced Imitation Learning.*
Mert Albaba, Sammy Christen, Christoph Gebhardt, **Thomas Langerak**, Michael J. Black, and Otmar Hilliges. 2024. 7th Robot Learning Workshop: Towards Robots with Human-Level Abilities (ICLR)

[A.2]: *Generalizing User Models through Hybrid Hierarchical Control.*
Thomas Langerak, Sammy Christen, Anna Feit and Otmar Hilliges. 2021. In Reinforcement Learning for Humans, Computer, and Interaction (CHI 2021 Workshop).

[A.3]: *A Demonstration on Dynamic Drawing Guidance via Electromagnetic Haptic Feedback.*
Thomas Langerak, Juan Zarate, Velko Vechev, Daniele Panozzo, and Otmar Hilliges. 2019. In The Adjunct Publication of the 32nd Annual ACM Symposium on User Interface Software and Technology.

[A.4]: *Aalto Interface Metrics (AIM): A Service and Codebase for Computational GUI Evaluation.*
Antti Oulasvirta, Samuli De Pascale, Janin Koch, **Thomas Langerak**, et al. 2018. In The 31st Annual ACM Symposium on User Interface Software and Technology Adjunct Proceedings.

Patents

[P.1]: Mental Model Aware Explainable Artificial Intelligence For Intelligent User Interfaces
US Patent 20250123860

TEACHING

(Co)-Lecturer

Computational Design and Interaction (120, M.Sc. + PhD) *2025*
Seminar on Human-AI Interaction (20, M.Sc. + PhD) *2025*
Seminar on Computational Haptics (20, M.Sc. + PhD) *2020, 2021*

Individual Lectures

Human-Computer Interaction: Computational Rationality (300, B.Sc.) *2023*
Human-Computer Interaction: (Computational) Haptics (100, B.Sc.) *2021-2023*
Human-Computer Interaction: Combinatorial Optimization (100, B.Sc.) *2021, 2022*
Human-Computer Interaction: Combinatorial Optimization (30, Industry) *2020*

Teaching Assistant

Seminar on Human Performance Capture (20, M.Sc. + PhD) *2024*
Computer Science I (>600, B.Sc.) *2022*
Ubiquitous Computing (20, B.Sc.) *2020, 2021*
Seminar on Advanced topics in Technical HCI (20, M.Sc. + PhD) *2020, 2021*
Human-Computer Interaction (100-300, B.Sc.) *2020-2023*
Seminar in Computational Interaction (20, M.Sc. + PhD) *2019*
Fairness, Equality and Accountability in Machine Learning (25, M.Sc.) *2019*

COMMUNITY SERVICE & ORGANIZING COMMITTEES

Associate Chair *2025*
Conference on Human Factors in Computing Systems (CHI)
Conference on Intelligent User Interface (IUI)

Co-Chair, Data *2021, 2022*
Symposium on User Interfaces, Systems and Technologies (UIST)

- Combine the different systems a conference uses and manage all data streams.
- Collaborate with different chairs.

Co-Chair, Virtual Experiences and Operations *2020*
Symposium on User Interfaces, Systems and Technologies (UIST)

- Transitioning a physical conference into virtual-only.
- Investigate the needs and solutions for virtual conferences.
- Manage all data streams.

REVIEWING

I review for all HCI conferences, including CHI, UIST, IUI, EICS and NordiCHI. I also review for journals such as Transactions on Haptics, and IEEE Sensors. Futhermore, I review for ML conferences (e.g., ICLR). Occasionally, I review for robotics such as HRI.

STUDENT SUPERVISION

Antti Immonen 2025
Master Thesis
Instance-level Contrastive Learning Methods for User Interfaces

Renate Zhang 2025
Summer Internship

Bayesian Optimization for Prototyping Interactive Devices
Under Review

Morgan Wang 2025
Research Assistant
Bayesian Optimization for Prototyping Interactive Devices
Under Review

Yugdeep Bangar 2024
Master Thesis
User Interface Optimization for the Quantified Self
Co-supervised with Prof. Alan Hanjalic, TU Delft

Caroline Sauget 2022
Master Thesis
Deep Reinforcement Learning for Sustainability

Mengfan Wu 2021
Master Thesis
Electromagnetic Tracking via Deep Learning
Published in *IEEE Transactions on Magnetics*

Aline Abler 2021
Master Thesis
Building a Hedgehog Pin Array Haptic Interface
Published in *World Haptics Conference 2021*
Honorable Mention for Best Paper

AWARDS

Honorable Mention - World Haptics Conference *2020*
NASA Europa Challenge Finalist *2019*
EIT Digital Excellence Scholarship *2019*

Grants

SIGCHI Diversity Grant (5000 Euros) *2020*
To enable people who otherwise cannot attend UIST to join.

INVITED TALKS

07/2025 Sorbonne University
Planning with Human Decision Models in Intelligent Systems

07/2025 Summer School on Computational Interaction
Reinforcement Learning for HCI

04/2024 TU Delft
User Agency and System Automation in Interactive Intelligent Systems

04/2024 TU Eindhoven
User Agency and System Automation in Interactive Intelligent Systems

12/2023 CMU, HCII
User Agency and System Automation in Interactive Intelligent Systems

11/2022 CMU, Augmented Perception Lab
Multi-Agent Reinforcement Learning for Adaptive UIs

References

Prof. Antti Oulasvirta, *Aalto University*, antti.oulasvirta@aalto.fi
Prof. Christian Holz, *ETH Żurich*, christian.holz@ethz.ch
Prof. Per Ola Kristensson, *University of Cambridge*, pok21@cam.ac.uk
Prof. David Lindlbauer, *CMU*, davidlindlbauer@cmu.edu